Wong Yeow Kang

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PROFILE SUMMARY

AI / Software Engineer who builds and ships ML systems end-to-end. Currently **building causal ML systems** over large-scale user data at TikTok Shop, and leading an LLM-based document processing system for an SME client **chaining vision-language model endpoints to schema validation**. Prior experience at a startup in Silicon Valley and strong data structures and algorithms foundation. Particularly **interested in LLMs**, including RAG, agents and model evaluation.

EDUCATION

NATIONAL UNIVERSITY OF SINGAPORE, SCHOOL OF COMPUTING Aug 2023 – May 2027
Bachelor of Science in Business Analytics (Honours)

- Grade Point Average: 4.47/5.0
- Specialisations: Machine Learning, Financial Analytics
- Activities: **NUS Overseas Colleges (Silicon Valley)**, Google Developer Student Club, NUS FinTech Society
- Relevant Coursework: Data Structures and Algorithms, Feature Engineering for Machine Learning, Fraud Analytics, Machine Learning, Linear Algebra and Advanced Calculus, Econometrics Modelling for Business Analytics, Programming Methodology (Python and Java)

STANFORD UNIVERSITY Aug 2025 – Dec 2025
Major in Entrepreneurship

- Grade: A
- Activities: **NUS Overseas Colleges (Silicon Valley)**

WORK EXPERIENCE

TIKTOK SHOP SINGAPORE *Machine Learning Engineer Intern* Aug 2026 – Present

- Build uplift models that allocate promotional incentive budget across users in four Southeast Asian markets.
- Found that production training labels counted cash-on-delivery purchases as non-purchases, missing more than half of true positives; rebuilt the daily sample pipeline with a 7-day label resolution step, raising held-out purchase AUC from 0.70 to 0.80 and spend and cost ranking AUC by 4%.
- Diagnosed selection bias from delayed labels that broke a continuously trained model in a live A/B test, inflating purchase predictions to 84–95% for every user; added a training-data guard and redeployed within 10 hours, cutting lowest-tier bonus decisions from 90% to 22%.
- Built the offline evaluation framework that decides which models go to live testing, including a 13-model feature ablation and a held-out ground-truth label set; the results stopped a planned 366-feature expansion.

WORKATO HQ PALO ALTO (NOC SILICON VALLEY) *Partner Solutions Consultant Intern* Aug 2025 – Jul 2026

- Built the partner submission and publishing pipeline for the Workato AI Marketplace from zero to launch, where partners list AI agents, MCP servers and accelerators; 27+ partners submitted in the first month, and automated review cut turnaround from 1 week to at most 2 days.
- Built a document parsing pipeline for a Brazilian bank on Workato's AI capabilities and exposed it as an MCP server callable from Claude Enterprise; reached 83.6% recall on a stratified test set, with per-class reporting to account for class imbalance.
- Built a RAG pipeline in LangChain for a Series C legal tech startup's FAQ assistant, chunking 53 FAQ documents and reranking the top-k results with a cross-encoder; evaluated on an 84-question set using MRR, MAP, Precision@k, Recall@k and faithfulness.
- Built the data pipelines and automated reporting behind partner lifecycle KPIs, replacing manual tracking with one source of truth used for ROI decisions.

MARIBANK *Data Analyst Intern (Consumer & SME)* Jan 2025 – Apr 2025

- Analysed month-over-month SMS engagement across the bank's consumer and SME base in MySQL and Excel, isolating the drivers of low-value sends and presenting a recommendation that cut campaign operating cost 73% from Dec'24 to Jan'25.
- Automated the monthly analysis and notification pipeline in Google Apps Script, removing all manual data updates and cutting irrelevant sends by over 40%.

NUS COMPUTING *Teaching Assistant (CS1010A Programming Methodology)* Aug 2024 – Dec 2024

- Taught data structures, algorithms, sorting and OOP in Python to first-year NUS Computing students.
- Achieved an 80% distinction rate for the class by delivering targeted support and engaging students in revision exercises.

PROJECTS

NUS BT4103 Capstone Project, Financial Document Processing Aug 2026 – Present

A VLM-based document extraction and claims management system built for a local precision-engineering SME.

- Building a multi-step pipeline on vision-language models (VLM) that reads invoices, receipts and claim forms directly from scanned images and PDFs into structured records, benchmarking Anthropic model endpoints against Hugging Face models on extraction accuracy and cost before production rollout.
- Building the validation and routing layer around model output; schema validation on extracted fields, error paths for malformed responses and rule-based approval routing; keeping downstream behaviour deterministic despite non-deterministic model output.
- Implementing the backend on FastAPI and Postgres, wiring model endpoints to database writes and downstream services; deployed on Vercel with CI.

Second Brain, Personal AI Agent System Jul 2026 – Present

A personal knowledge base in Obsidian, run by Claude Code agents and queried by a fine-tuned language model.

- Built an agent system in Claude Code on top of an Obsidian vault, using custom skills and MCP connectors to find relevant events, track my stock portfolio with recommendations, and triage my email.
- Designed the vault as shared memory for the agents: each runs on a schedule, reads context from the same notes and writes its results back, so every output reflects up-to-date personal context.
- Fine-tuning an open-weight language model on the vault's notes so I can query it about anything I've recorded in natural language.

SpotiFind, Music Recommendation System Jan 2025 – May 2025

A machine learning-based music recommender using audio features, lyric sentiment, and hybrid filtering.

- Developed a machine learning-based recommendation system leveraging acoustic features, harmonic pattern analysis and BERT-based sentiment analysis for contextual relevance in lyrics.
- Engineered features from a dataset of over 250,000 songs, applying K-means clustering to recover latent genre structure in a high-dimensional feature space.
- Applied collaborative filtering and content-based models, analysing spectrogram data with cosine similarity to recommend tracks with similar frequency distributions.

CO-CURRICULAR ACTIVITIES

NUS FINTECH SOCIETY, Machine Learning Analyst Trainee Aug 2024 – Dec 2024

- Utilised Python libraries such as Pandas and NumPy for data analytics and visualization.
- Built end-to-end ML pipelines in scikit-learn spanning preprocessing, feature engineering and out-of-sample model evaluation.

GOOGLE DEVELOPER STUDENT CLUB (NUS), Business Associate Aug 2023 – May 2024

- Collaborated with a team of undergraduate developers and UI/UX designers to create Project Ignite, to streamline volunteer recruitment and management processes for non-profit startups, driving social impact and community engagement.
- Established application requirements and features for the Volunteer Management System (VMS).
- Designed the User-Profile Subsystem for the Volunteer Management System (VMS) platform, ensuring a seamless user experience for both non-profit organizations and volunteers.

TECHNICAL SKILLS AND PROFESSIONAL CERTIFICATIONS

- Written & Spoken Languages: English, Mandarin
- Programming Languages: Python, Java, R, SQL, Javascript, C++ (in progress)
- AI / Machine Learning: PyTorch, scikit-learn, NumPy, Hugging Face Transformers, LLMs, Spark, causal inference and uplift modelling, recommendation systems, VLM
- Web & Tooling: Fast API, Vue, ReactJS, Flask, Node.js, Git, Postgres, CI/CD
- LLMs (In Progress): Tokenization & Embeddings, Transformers, SFT & LoRA Fine-tuning, RLHF, PPO, DPO, GRPO, RAG, ReAct Agents, Tool Calling, LLM Evaluation
- Professional Certifications: CS50AI Introduction to AI with Python, Google Advanced Data Analytics Certification, Quantitative Risk Management in Python